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DEGLI STUDI
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Entrepreneurship & Innovation
Statistics for management
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Electrical Equipment manufacturing in Italy

(Lombardia – Lazio – Puglia)

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1. Electrical Equipment Manufacturing:

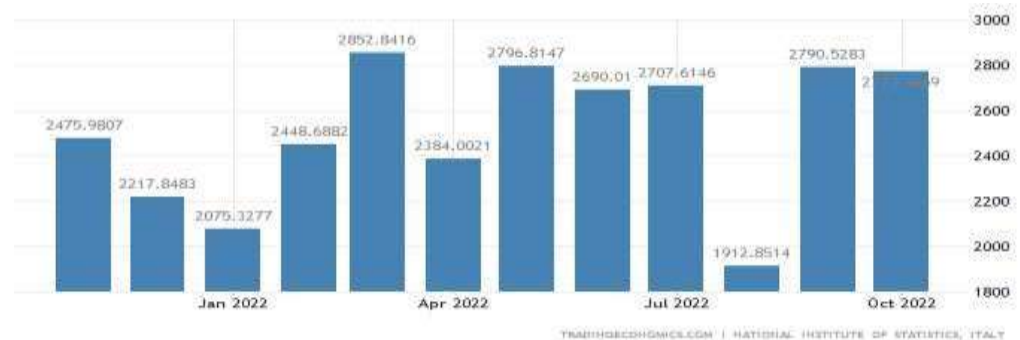
The electrical equipment market consists of sales of products that generate, distribute and use electrical power by entities (organizations, sole traders, and partnerships) which produce products that generate, distribute and use electrical power.

The main types of electrical equipment include electric lighting equipment, household appliances, and power generation and transmission equipment, among others. Household appliances are used for functions such as cooking, cleaning, and food storage. The equipment can be purchased through OEM and aftermarket channels by both B2B and B2C end users. The global electrical equipment market experienced a 7.6% compound annual growth rate from 2021 to 2022, reaching \$1,490.12 billion. The market is expected to continue growing, reaching \$1,943.24 billion by 2026 with a CAGR of 6.9%. However, the Russia-Ukraine war has disrupted the market, causing economic sanctions, a surge in commodity prices, and supply chain disruptions that have affected many markets globally.

In **Italy** exports of electrical equipment decreased to 2774.47 EUR Million in October from 2790.53 EUR Million in September of 2022.

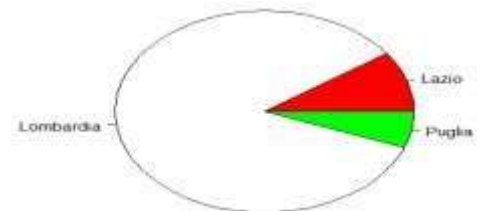
Analysts are most optimistic on the heavy electrical equipment industry, expecting annual earnings

growth of 26% over the next 5 years. In contrast, the Electrical Components industry is expected to see its earnings growth to stay flat over the next few years.



2. Our Database

Our database is imported from **Orbis website**, so we are going to analyze a sample of companies specialize in manufacturing of electrical equipment from three different regions in Italy (Lombradia-Lazio-Puglia), and we have selected only the **active companies**, so we excluded the inactive companies and companies with unknown situation. Totally our observations are **1509** in **2021** within 18 variables .We just use a simple table that present the number of companies that belongs to each region separately.



Lombradia	Lazio	Puglia
1273	148	88

3. Research proposal:

As sales is one of the most important matters that companies want to amplify, so our question is **what are the most influential factors on sales?** So we are going to get a suitable answer through multiple linear regression method. **Besides**, after getting an answer we are going to use cluster (k-means) method to analyze **how the companies distribution looks like when divided into subgroups in term of their branches and profits?**

4. Multiple linear regression (As the first statistical method):

In order to simplify our work in R we renamed the variables as indicated in the following table so (S) is a shorthand of sales and so on.

Sales	Number of Employee	Financial Expenses	Total Assets	Number of branches	Taxation	Cash flow
s	E	EX	A	B	TX	CS

call:

```
lm(formula = facts$S ~ facts$E + facts$EX + facts$A + facts$B +  
    facts$TX + facts$CS)
```

The multiple linear regression formula that consists of sales as the **dependent** variable and number of employees, total financial expenses, total assets, number of branches, amount of taxation and amount of cash flow as the **predictors** results in the following table:

```
-----  
Dependent variable:  
-----  
S  
-----  
E          392.066***  
          (12.335)  
EX         -31.323***  
          (1.688)  
A           0.430***  
          (0.022)  
B          1,954.659***  
          (677.370)  
TX           7.875***  
          (1.932)  
CS          -3.144***  
          (0.243)  
Constant   -5,143.243***  
          (1,032.269)  
-----  
Observations      1,509  
R2                0.833  
Adjusted R2       0.832  
Residual Std. Error 36,285.770 (df = 1502)  
F Statistic      1,245.508*** (df = 6; 1502)  
-----  
Note:             *p<0.1; **p<0.05; ***p<0.01  
└───┘
```

The **general overview** of these results is:

Adjusted R-squared value of 0.832 means that **(83%)** of our response variable (sales) can be explained by our model.

F-statistic is 1246 and **p-value is very small** (less than $2.2e-16$) indicating that the model is significantly better than the null model. And **lower P-value** (less than 0.05) means that **null hypothesis** is rejected, in our case all predictor variables are significantly different from zero. In addition, there are some **important indications** of the relationship between the response variable and the predictors:

- a-** The slope of the predictor number of branches is the most influencing factor on sales which mean that a one unit increase in Branch leads to approximately 1.954 increase in sales (USD).
- b-** Also of the predictor number of employees significantly affects sales so one unit increase in number of employees results in increasing in sales with (392) units. In this case we can predict especially the number of marketers that perform the main role in business cycle.
- c-** In terms of negative relationships: We can see that the predictor (total expenses) has a negative effect on sales, means that increasing one unit in this predictor leads to declining in sales with (-31.3) units. Similarly, to the relationship between cash flow as a predictor and sales as response but this predictor has a less negative effect means that increasing (cash flow) with one-unit results in decreasing in sales with (-3) units.

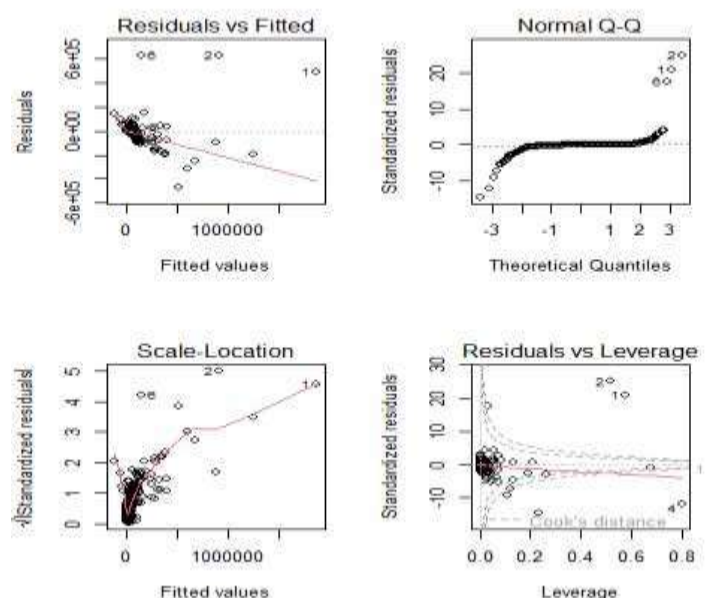
So the formula of our multiple linear regression of our model can be written as following:

$$y = \beta_0 + \beta_1x_1 + \beta_2x_2 + \dots + \beta_p x_p$$

$$S = -5143.2431 - 31.3230EX + 0.4298A + 1954.6592B + 7.8749TX - 3.1436CS + 392.0657E$$

In order to better understanding of our model we considered these **plots**:

The residual analysis involves examining plots of **the residuals vs. fitted** (the difference between the observed and predicted values) to assess the quality of the linear regression model. In a good linear regression model, the residuals should follow a linear pattern and a normal distribution. In the current model, the residuals do not follow a linear



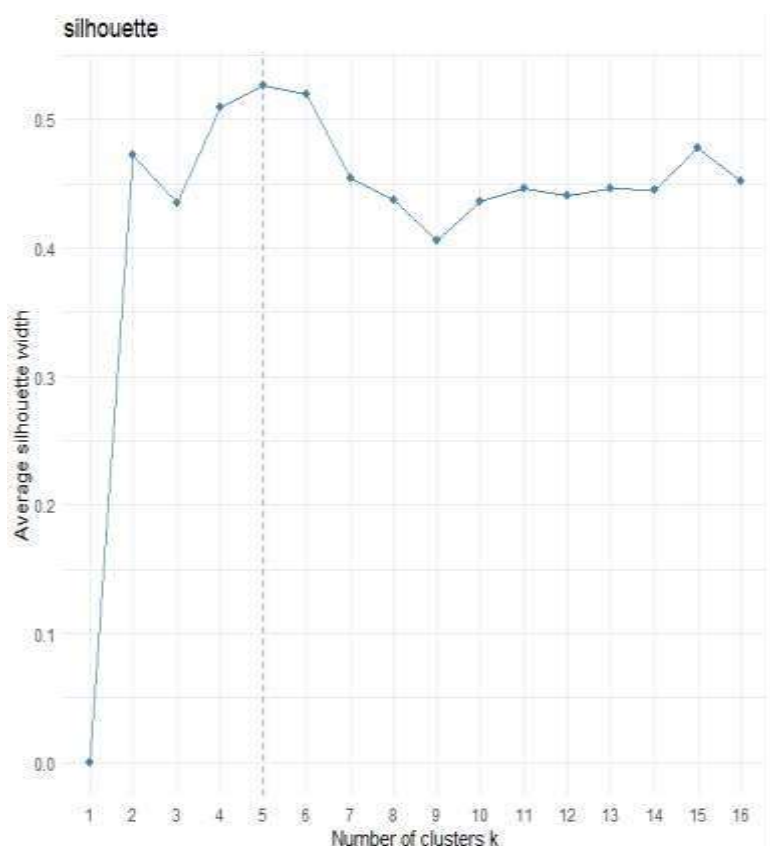
pattern, and the normal **Q-Q** indicates outliers in the distribution. The **scale-location** plot indicates that the variance of the residuals is consistent, there is some residual error terms that could potentially affect the regression analysis that indicated in **residual vs leverage**.

5. Clustering (as our second statistical method):

-Through cluster we seek to group our data in distinct clusters or groups. We are going to work using **K-means** method as a popular and prior by iteratively reassigning data points to the nearest cluster based on the mean values of the variables in each cluster. We depend on two variables to make this cluster the first is "**Profit margin**" And the second is "**Number of branches**".

- **Profit margin** is a major factor in measuring the profitability and the financial health of a business by dividing the net profit of a business by its revenue, so it is a good indicator for investors that helps them to evaluate the efficiency and effectiveness of the company's operations. Also, **number of branches** in our case has played an important role in sales volume so we consider that it is a good subject in clustering our data.

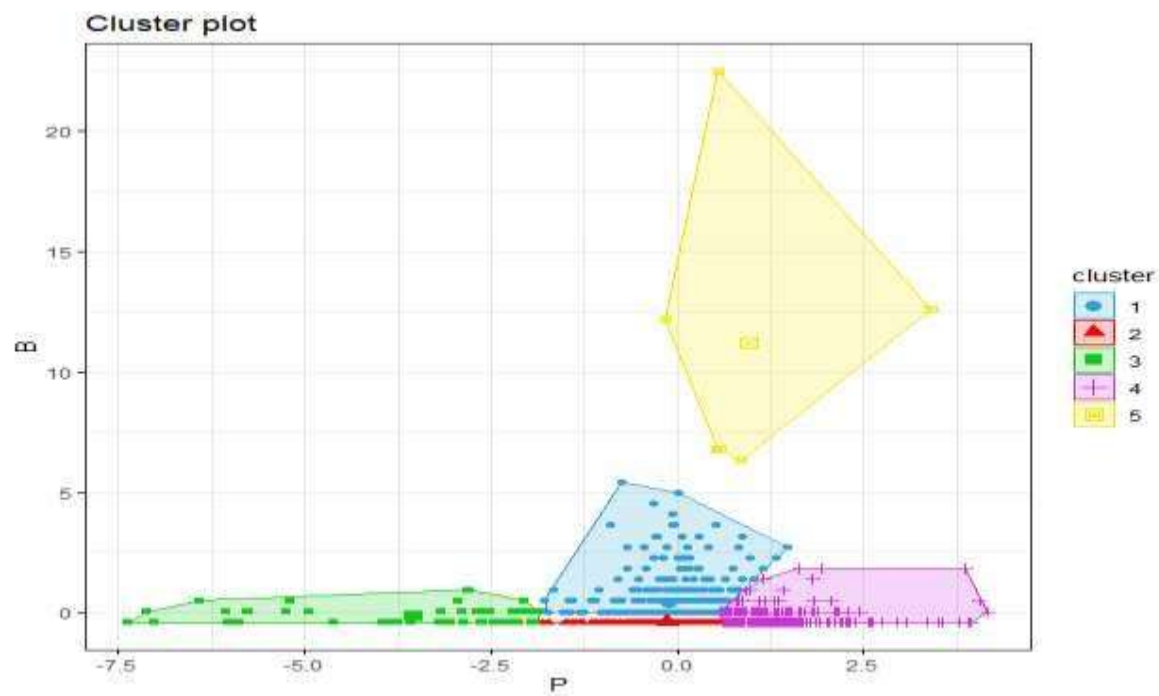
-In order to get the appropriate number of clusters for our dataset we have chosen "**Silhouette**" method to compute the average silhouette width for each possible number of clusters, the number of cluster that results in the highest average silhouette width is generally considered the best number of clusters (indicates the highest degree of similarity within clusters and dissimilarity between clusters).In our case according to Silhouette's plot the optimum number of clusters is **(5)**, as it can be seen .



- In order to analyze the results of cluster we make table for our results and plot it:

1	2	3	4	5
577	636	49	241	6
>				

Cluster means:			
	P	B	
1	-0.1159137	0.3982956	
2	-0.1321469	-0.4029539	
3	-3.5343077	-0.1375625	
4	1.3209836	-0.1406003	
5	0.9586129	11.1812272	



As **B** the (Y – axis) stands for **Number of branches** and **P** the (x-axis) stands for **profit margin**. We can infer from this plot of 1509 observations that **cluster 4** has the highest profit, while **cluster 5** has the highest number of branches. **Cluster 3** has the lowest profit and a very low number of branches. **Cluster 1** is the middle of clusters, come just after cluster 5 in term of branches and after cluster 5 on the profit level. **Meanwhile cluster 2 is neutral** on both sides representing companies with 0 branches and a profit significantly close to 0.

Conclusion:

The linear regression model gave us an important indicator related to sales that number of Branches has a significant role in order to Amplify sales, the formula can be written $S = -5143.2431 + 1954.6592B$. Besides, cluster method provided us with a good understanding of the different groups of the data indicates that a huge proportion of the data has less than 2 branches. To sum up, companies that don't have branches should consider creating some of it, because of the major impact it has on the sales. And the optimal way on doing it is having a minimum of 5 branches in order to secure a profit higher than 0.

The outliers seen in Q-Q plot in multiple regression can be related to the outliers seen in clustering plot (cluster number 5) which is represented by companies that have more than 5 branches (about 6 companies)

Additionally, companies with less than 2 branches should consider expanding their branch network in order to increase their sales. This can be achieved by analyzing the data and identifying the optimal number of branches to have for maximum sales. Furthermore, it is important to note that the linear regression model and cluster method used in this study can provide valuable insights for companies looking to improve their sales performance. Overall, the results suggest that having a sufficient number of branches is crucial for amplifying sales and securing a profit.